

机器人学

第六章：操作臂（机械臂）的 轨迹规划

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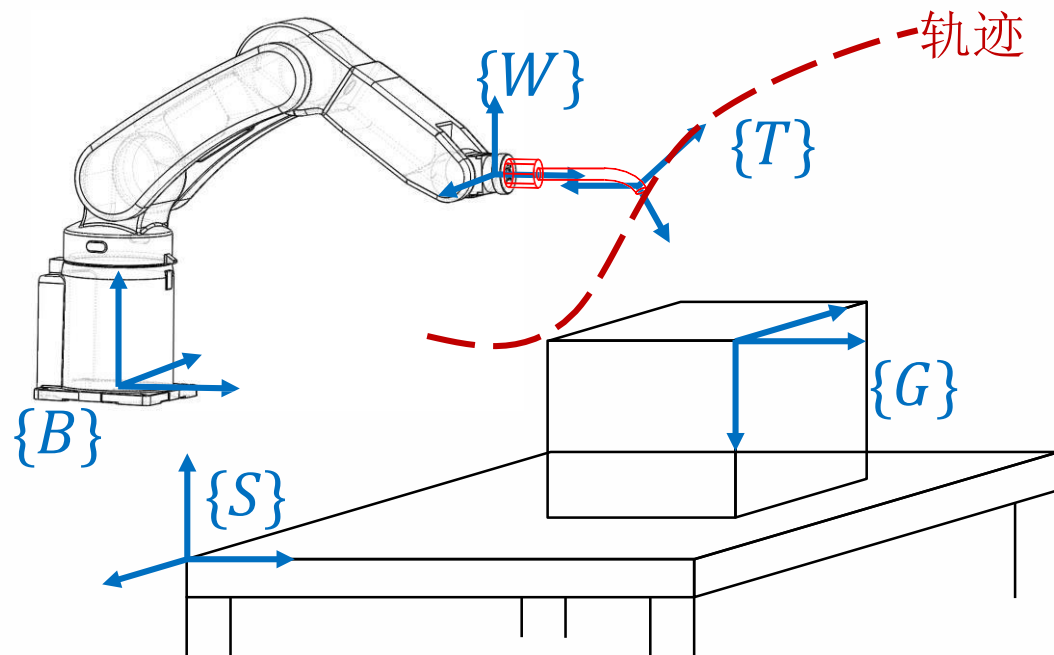


引言 -1

□ 轨迹：机械手臂（的末端点或操作点）的位置、速度、加速对时间的历程

□ 可进一步定义成 $\{T\}$ 相对 $\{G\}$ 的状态历程

和手臂种类无关， $\{G\}$ 也可随时间变动（如输送带）



引言 -2

- 对象取放任务中手臂末端点（或物件）的运动历程，即是轨迹规划所进行的内容
 - ◆ 例如：
 - ◆ 机械手臂夹住放在桌上的杯子，移动手臂将杯子挂到墙上的杯架

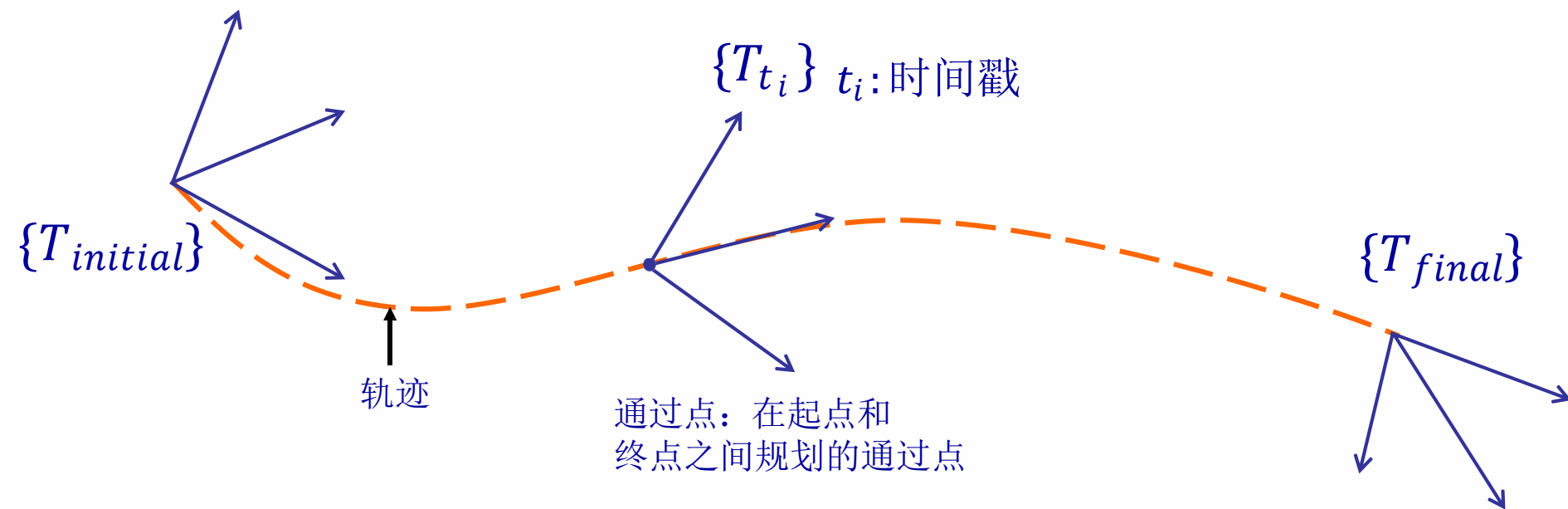


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- ⑧ https://www.bilibili.com/video/BV1tv411a77n?spm_id_from=333.999.0.0
 - ⑧ https://www.bilibili.com/video/BV19K411A7kj?spm_id_from=333.999.0.0
 - ⑧ https://www.bilibili.com/video/BV1N3411x7QK?spm_id_from=333.999.0.0

引言 -3

- 理想轨迹：平滑路径（即，连续一阶导数）

常常为 ζT



关节空间下的轨迹规划 -1

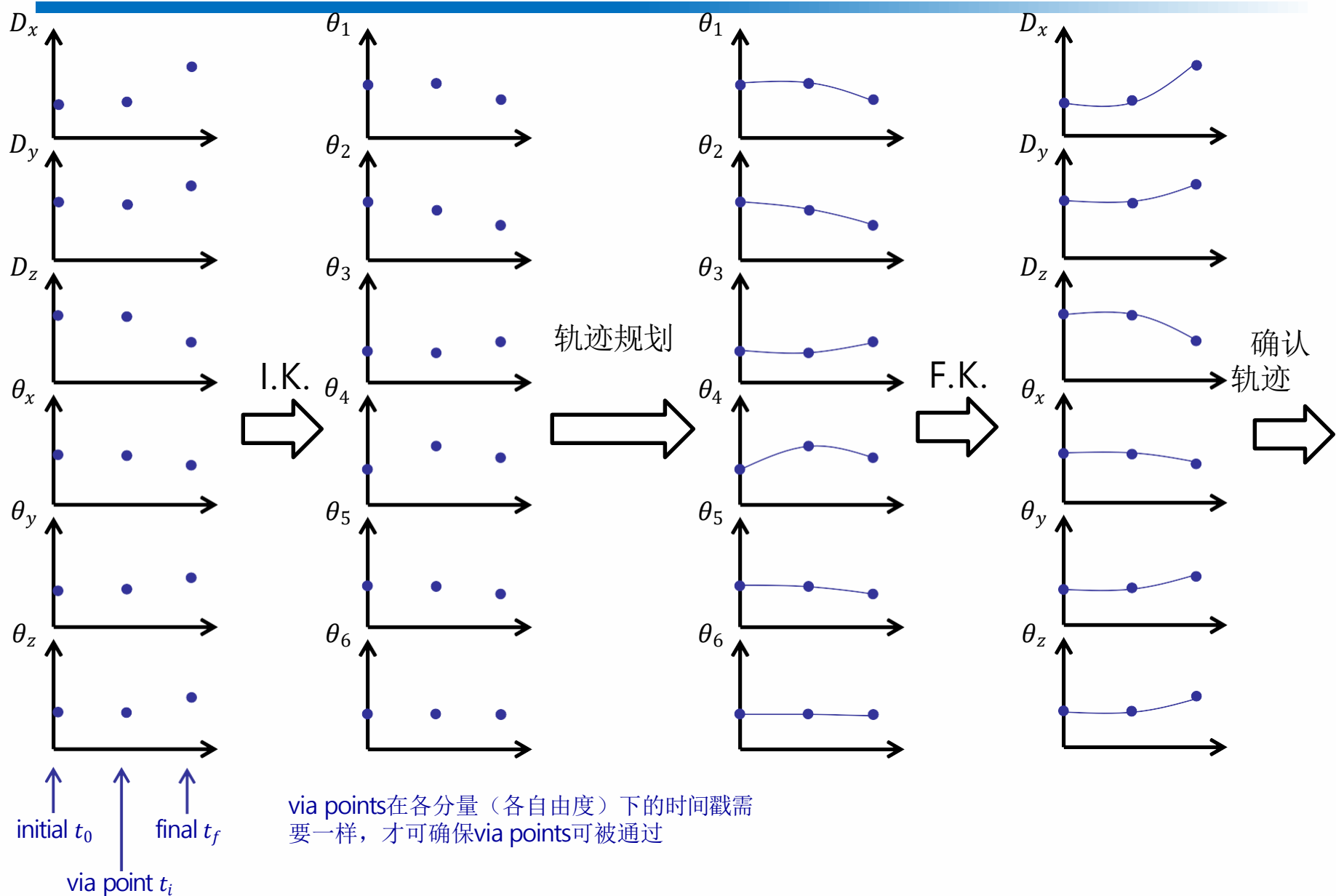
□ 步骤

- ◆ 定义{T}相对于{G}的initial, via, & final points, ${}^G T_i$
(包含移动和转动自由度)
 - i=1 initial
 - i=2~N-1 via points
 - i=N+1 final

将 ${}^G T_i$ 以6个参数方式来表达 ${}^G X_T = \begin{bmatrix} {}^G P_{T \text{ org}} \\ \text{ROT}({}^G \hat{K}_T, \theta) \end{bmatrix}$
非旋转矩阵的方式表达

- ◆ 逆向运动学将手臂末端点状态转换到关节状态: ${}^G X_T \rightarrow \theta_i$
- ◆ 对所有关节规划平滑轨迹
- ◆ 正向运动学:将关节状态转换到手臂末端点状态, 检查莫短点在笛卡尔空间下轨迹的可行性

关节空间下的轨迹规划 -2



笛卡尔空间下的轨迹规划 -1

□ 步骤

- ◆ 定义{T}相对于{G}的initial, via, & final points, ${}^G T_i$
(包含移动和转动自由度)
i=1 initial
i=2~N-1 via points
i=N+1 final

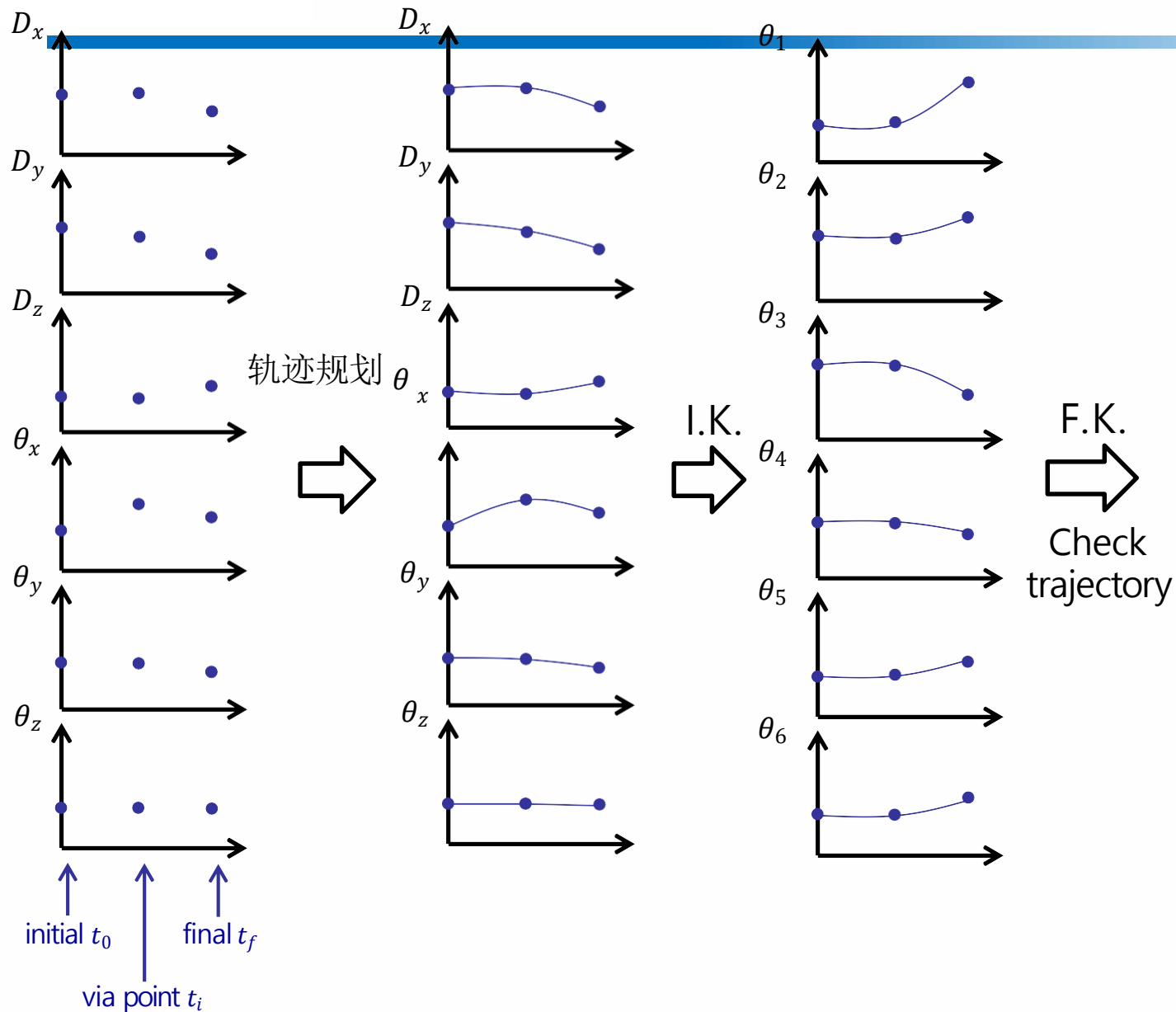
将 ${}^G T_i$ 以6个参数方式来表达 ${}^G X_T = \begin{bmatrix} {}^G P_{T \text{ org}} \\ ROT({}^G \hat{K}_T, \theta) \end{bmatrix}$

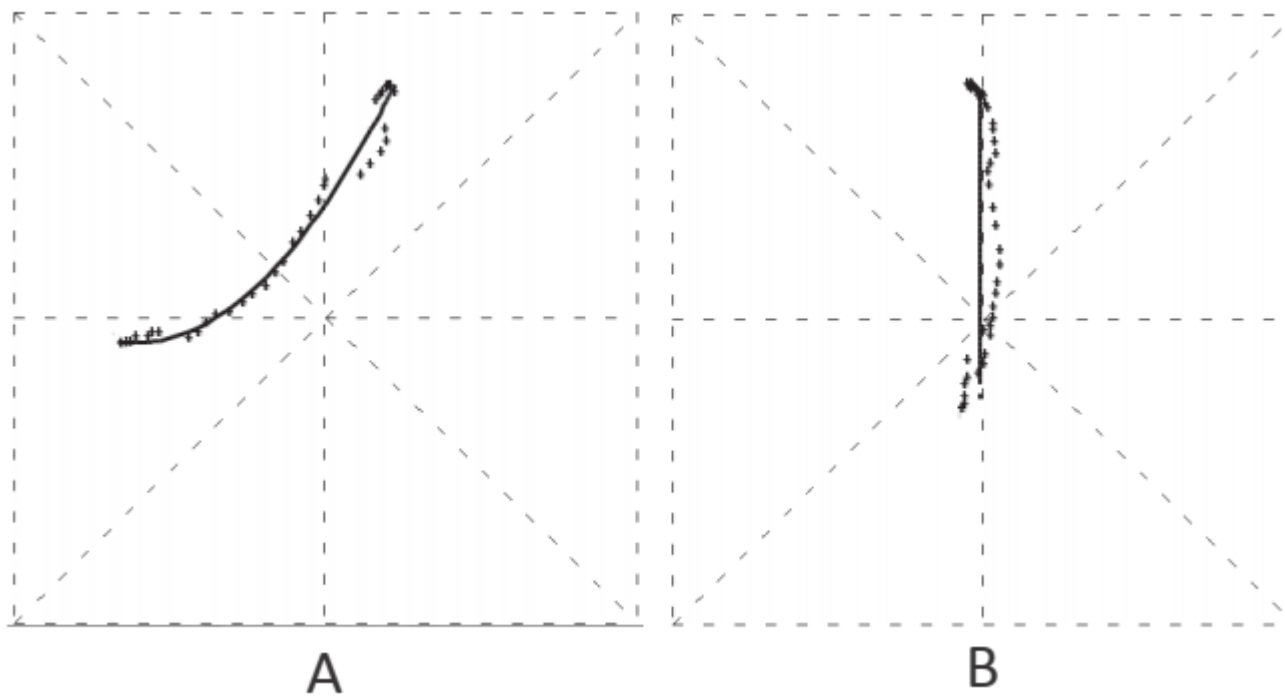
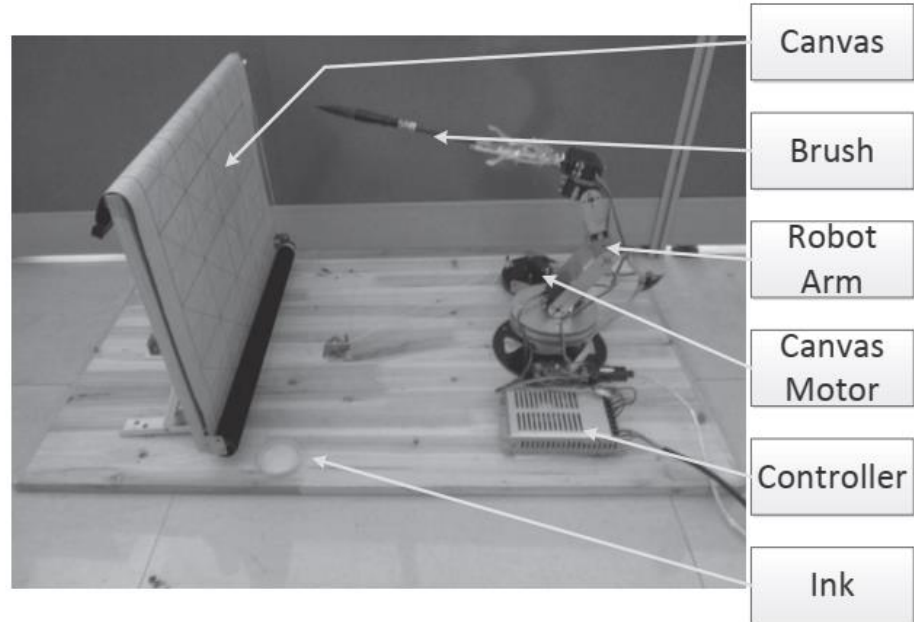
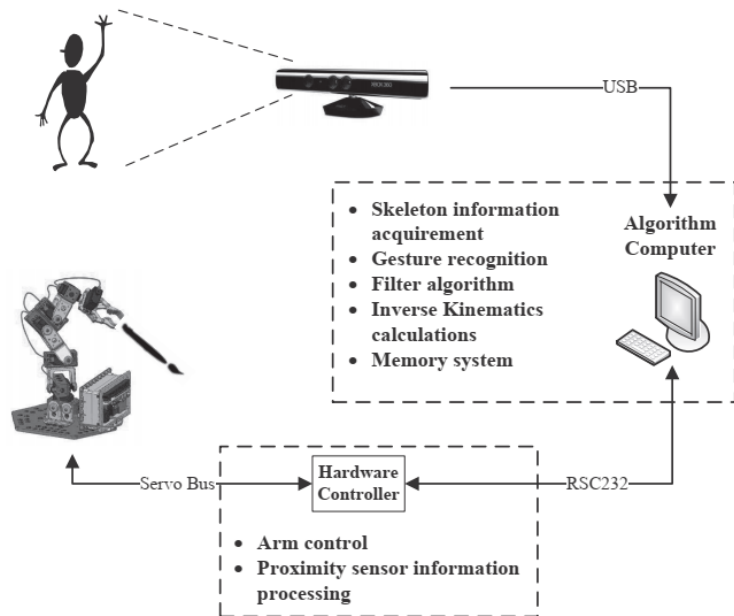
- ◆ 对所有手臂末端点状态规划平滑轨迹
- ◆ 将规划好手臂末端点状态的轨迹点转换到关节状态: ${}^G X_T \rightarrow \Theta_i$
- ◆ 检查关节状态在关节空间下轨迹的可行性

□ 注意:

- ◆ 较具物理直观的轨迹
- ◆ 较高的运算负载 (i.e., IK)

笛卡尔空间下的轨迹规划 -2

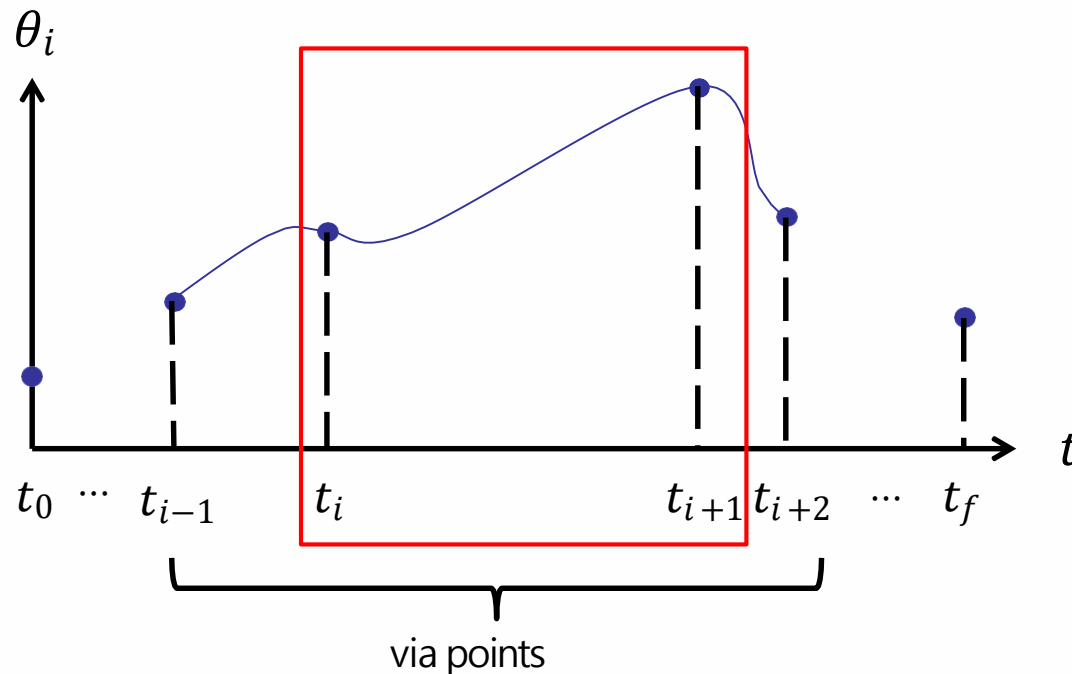




三次多项式说明 -1

□ 原则

- ◆ 轨迹: 不同轨迹区段 $[t_i \ t_{i+1}]$ 以不同参数的函数来规划
- ◆ 平滑: 需定义各函数的边界条件 (包含位置和速度
 $\theta(t_i), \theta(t_{i+1}), \theta'(t_i), \theta'(t_{i+1})$, 有4个条件)
- ◆ 以cubic polynomial (三次多项式) 来规划轨迹



- 1 linear
- 2 quadratic
- 3 cubic
- 4 quartic
- 5 quintic
- 6 hexic (sextic)
- 7 heptic (septic)
- 8 octic
- 9 nonic
- 10 decic

三次多项式说明-2

□ 解cubic polynomial

◆ 通式

$$\theta(\tilde{t}) = a_0 + a_1\tilde{t} + a_2\tilde{t}^2 + a_3\tilde{t}^3 \quad 4 \text{ unknowns: } a_j \quad j=0\sim 3$$

◆ 对每一个区段: $t \in [t_i, t_{i+1}]$

$$\tilde{t} = t - t_i \quad \text{so } \tilde{t}|_{t=t_i} = 0 \text{ and } \tilde{t}|_{t=t_{i+1}} \equiv \Delta t = t_{i+1} - t_i > 0$$

每一个区段 $[t_i, t_{i+1}]$ 的 Δt 可以不同，取决于via points的设置

边界条件

$$\theta(\tilde{t}|_{t=t_i}) = \theta_i = a_0 \quad \text{①}$$

$$\theta(\tilde{t}|_{t=t_{i+1}}) = \theta_{i+1} = a_0 + a_1\Delta t + a_2\Delta t^2 + a_3\Delta t^3 \quad \text{②}$$

$$\dot{\theta}(\tilde{t}|_{t=t_i}) = \dot{\theta}_i = a_1 \quad \text{③}$$

$$\dot{\theta}(\tilde{t}|_{t=t_{i+1}}) = \dot{\theta}_{i+1} = a_1 + 2a_2\Delta t + 3a_3\Delta t^2 \quad \text{④}$$

三次多项式说明-3

□ 解联立方程式 ① ② ③ ④

$$a_2 = \frac{3}{\Delta t^2}(\theta_{i+1} - \theta_i) - \frac{2}{\Delta t}\dot{\theta}_i - \frac{1}{\Delta t}\dot{\theta}_{i+1}$$

$$a_3 = -\frac{2}{\Delta t^3}(\theta_{i+1} - \theta_i) + \frac{1}{\Delta t^2}(\dot{\theta}_{i+1} + \dot{\theta}_i)$$

三次多项式说明-4

- 以Matrix operation方式运算

$$\begin{bmatrix} \theta_i \\ \theta_{i+1} \\ \dot{\theta}_i \\ \dot{\theta}_{i+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & \Delta t & \Delta t^2 & \Delta t^3 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2\Delta t & 3\Delta t^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\Theta = T_{4 \times 4}(\Delta t) \cdot A$$

$$\det(T_{4 \times 4}) = -\Delta t^4 \neq 0 \text{ as long as } \Delta t \neq 0$$

所以

$$A = T_{4 \times 4}^{-1} \cdot \Theta$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{3}{\Delta t^2} & \frac{3}{\Delta t^2} & -\frac{2}{\Delta t} & -\frac{1}{\Delta t} \\ \frac{2}{\Delta t^3} & -\frac{2}{\Delta t^2} & \frac{1}{\Delta t^2} & \frac{1}{\Delta t^2} \end{bmatrix} \begin{bmatrix} \theta_i \\ \theta_{i+1} \\ \dot{\theta}_i \\ \dot{\theta}_{i+1} \end{bmatrix}$$

多段三次多项式 -1

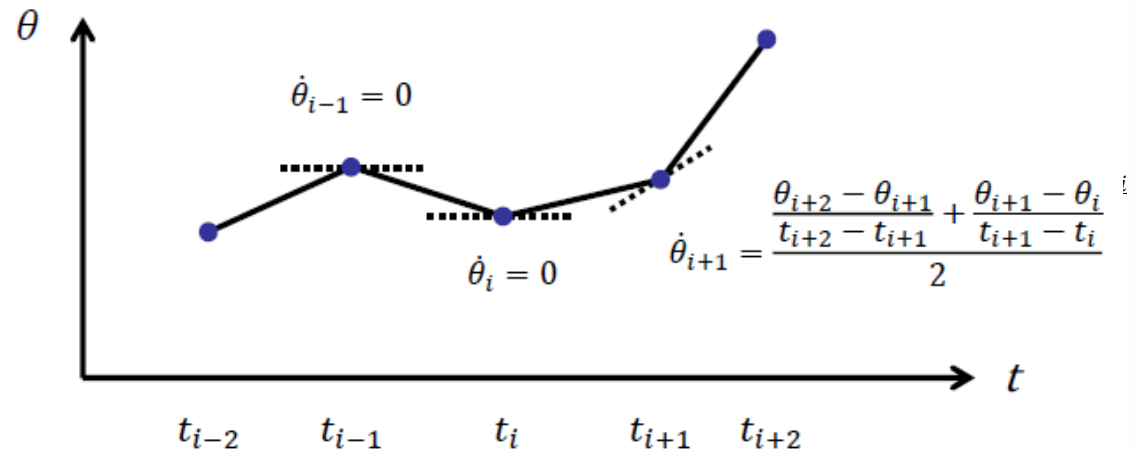
□ 如何选择速度条件 $\dot{\theta}_i$ 和 $\dot{\theta}_{i+1}$?

- ◆ 直接定义，不论在 Cartesian space 或 关节 space
不建议，过于复杂，尤其轨迹落在singular points附近时

- ◆ 自动生成

Ex: 如果 $\dot{\theta}_i$ 在 t_i 前后变号，选择 $\dot{\theta}_i = 0$

如果 $\dot{\theta}_i$ 在 t_i 前后同号，选择平均



以此两个方式，不同区段的三次多项式可以分开求解

多段三次多项式 -2

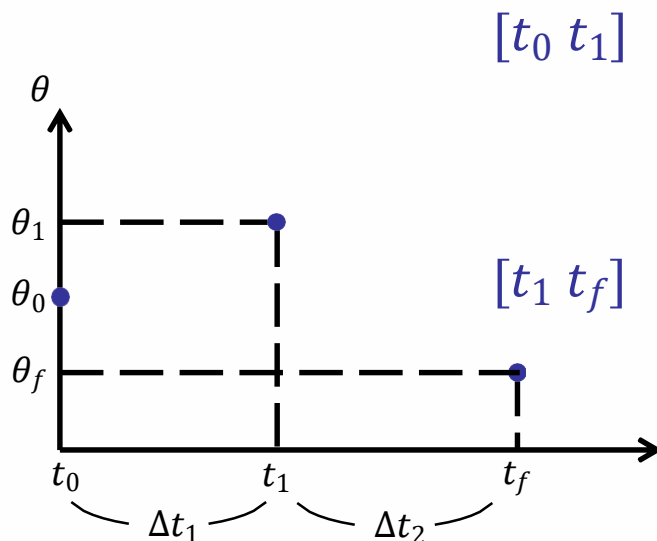
- ◆ 规划速度使过程中加速度也连续

(有效使用此可调控的变数)

不同区段的三次多项式需要整合一起，一并求解

- ◆ Example: A trajectory with one via point

各段在运算时将时间平移到0开始



$$\tilde{t} = t_1 - t_0 \quad [0 \ \Delta t_1]$$

$$\theta_I(\tilde{t}) = a_{10} + a_{11}\tilde{t} + a_{12}\tilde{t}^2 + a_{13}\tilde{t}^3$$

$$\tilde{t} = t_f - t_1 \quad [0 \ \Delta t_2]$$

$$\theta_{II}(\tilde{t}) = a_{20} + a_{21}\tilde{t} + a_{22}\tilde{t}^2 + a_{23}\tilde{t}^3$$

$\Rightarrow 8 \text{ unknowns}$

多段三次多项式 -3

- ◆ Example: A trajectory with one via point (cont.)

4 position B.C.s

2 for each θ (t) $j = \text{I, II}$

$$\begin{cases} \theta_0 = a_{10} \\ \theta_1 = a_{10} + a_{11}\Delta t_1 + a_{12}\Delta t_1^2 + a_{13}\Delta t_1^3 \\ \theta_1 = a_{20} \\ \theta_f = a_{20} + a_{21}\Delta t_2 + a_{22}\Delta t_2^2 + a_{23}\Delta t_2^3 \end{cases}$$

2 velocity B.C.s

$$\begin{cases} \dot{\theta}_0 = \boxed{0} = a_{11} \\ \dot{\theta}_f = \boxed{0} = a_{21} + 2a_{22}\Delta t_2 + 3a_{23}\Delta t_2^2 \end{cases}$$

not necessary "0"

Via point

velocity continuity

acceleration continuity

$$\begin{cases} \dot{\theta}_1 = a_{11} + 2a_{12}\Delta t_1 + 3a_{13}\Delta t_1^2 = a_{21} \\ \ddot{\theta}_1 = 2a_{12} + 6a_{13}\Delta t_1 = 2a_{22} \end{cases}$$

$\Rightarrow 8 \text{ equations}$

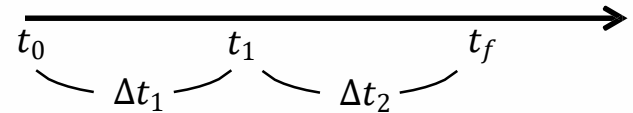
多段三次多项式 -4

- ◆ Example: A trajectory with one via point (cont.)

8 equations, 8 unknowns

代数解法

(when $\Delta t_1 = \Delta t_2 = \Delta t$)



$$a_{10} = \theta_0$$

$$a_{11} = 0$$

$$a_{12} = \frac{12\theta_1 - 3\theta_f - 9\theta_0}{4\Delta t^2}$$

$$a_{13} = \frac{-8\theta_1 + 3\theta_f + 5\theta_0}{4\Delta t^3}$$

$$a_{20} = \theta_1$$

$$a_{21} = \frac{3\theta_f - 3\theta_0}{4\Delta t}$$

$$a_{22} = \frac{-12\theta_1 + 6\theta_f + 6\theta_0}{4\Delta t^2}$$

$$a_{23} = \frac{8\theta_1 - 5\theta_f - 3\theta_0}{4\Delta t^3}$$

多段三次多项式 -5

- ◆ Example: A trajectory with one via point (cont.)

矩阵解法

$$[0 \ \Delta t_1] \quad \theta_I(\tilde{t}) = a_{10} + a_{11}\tilde{t} + a_{12}\tilde{t}^2 + a_{13}\tilde{t}^3$$

$$\Theta_{8 \times 1} = T_{8 \times 8} A_{8 \times 1}$$

$$[0 \ \Delta t_2] \quad \theta_{II}(\tilde{t}) = a_{20} + a_{21}\tilde{t} + a_{22}\tilde{t}^2 + a_{23}\tilde{t}^3$$

$$\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_1 \\ \theta_f \\ \dot{\theta}_0 \\ \dot{\theta}_f \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & \Delta t_1 & \Delta t_1^2 & \Delta t_1^3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \Delta t_2 & \Delta t_2^2 & \Delta t_2^3 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2\Delta t_2 & 3\Delta t_2^2 \\ 0 & 1 & 2\Delta t_1 & 3\Delta t_1^2 & 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 6\Delta t_1 & 0 & 0 & -2 & 0 \end{bmatrix} \begin{bmatrix} a_{10} \\ a_{11} \\ a_{12} \\ a_{13} \\ a_{20} \\ a_{21} \\ a_{22} \\ a_{23} \end{bmatrix}$$

$$A_{8 \times 1} = T_{8 \times 8}^{-1} \Theta_{8 \times 1}$$

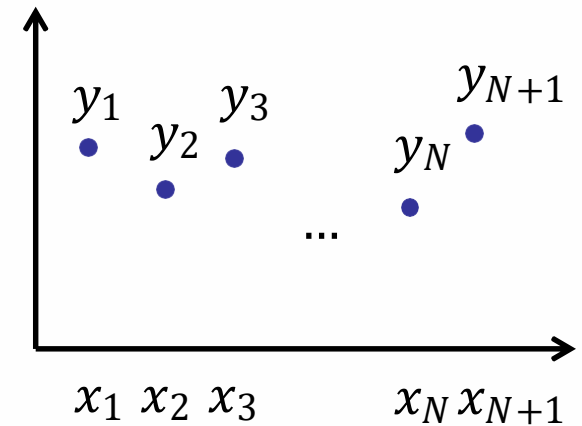
$$\det(T_{8 \times 8}) = 4\Delta t_1^4 \Delta t_2^3 + 4\Delta t_1^3 \Delta t_2^4$$

$$\neq 0 \text{ as long as } \Delta t_1 \neq 0, \Delta t_2 \neq 0, \Delta t_1 \neq -\Delta t_2$$

一般三次多项式 -1

□ 一般三次样条函数

N+1 set points (x_i, y) $\left\{ \begin{array}{l} 1 \\ N-1 \\ 1 \end{array} \right.$ $\begin{array}{l} \text{initial} \\ \text{via} \\ \text{final} \end{array}$



N cubic functions

$$s_j(x) = a_j + b_j x + c_j x^2 + d_j x^3$$

$$\begin{array}{l} x_j \leq x \leq x_{j+1} \\ j = 1 \dots N \end{array}$$

⇒ total 4N unknown 系数

一般三次多项式 -2

Position conditions at both ends of each $s_j(x)$

$\Rightarrow 2N$ conditions

Velocity & acceleration continuity conditions at via points

$\Rightarrow 2(N-1)$ conditions

还需要 2 CONDITIONS 以求解

Revisit example: A trajectory with one via point

$$\begin{cases} y_1 = s_1(x_1) \\ y_2 = s_1(x_2) \end{cases} \quad \begin{cases} y_2 = s_2(x_2) \\ y_3 = s_2(x_3) \end{cases}$$

$$\dot{y}_2 = \dot{s}_1(x_2) = \dot{s}_2(x_2)$$

$$\ddot{y}_2 = \ddot{s}_1(x_2) = \ddot{s}_2(x_2)$$

共 6 conditions

一般三次多项式 -3

最后 2 conditions 的选择方法

$$(1) \quad \ddot{s}_1(x_1) = \ddot{s}_N(x_{N+1}) = 0$$

定义加速度, Natural cubic spline

$$(2) \quad \dot{s}_1(x_1) = u \quad \dot{s}_N(x_{N+1}) = v$$

定义速度, Clamped cubic spline

$$(3) \quad \begin{array}{l} \text{if } s_1(x_1) = s_N(x_{N+1}) \\ \text{use } \dot{s}_1(x_1) = \dot{s}_N(x_{N+1}) \\ \ddot{s}_1(x_1) = \ddot{s}_N(x_{N+1}) \end{array}$$

周期运动的连续性, Periodic cubic spline

示例：RRR机械臂-1

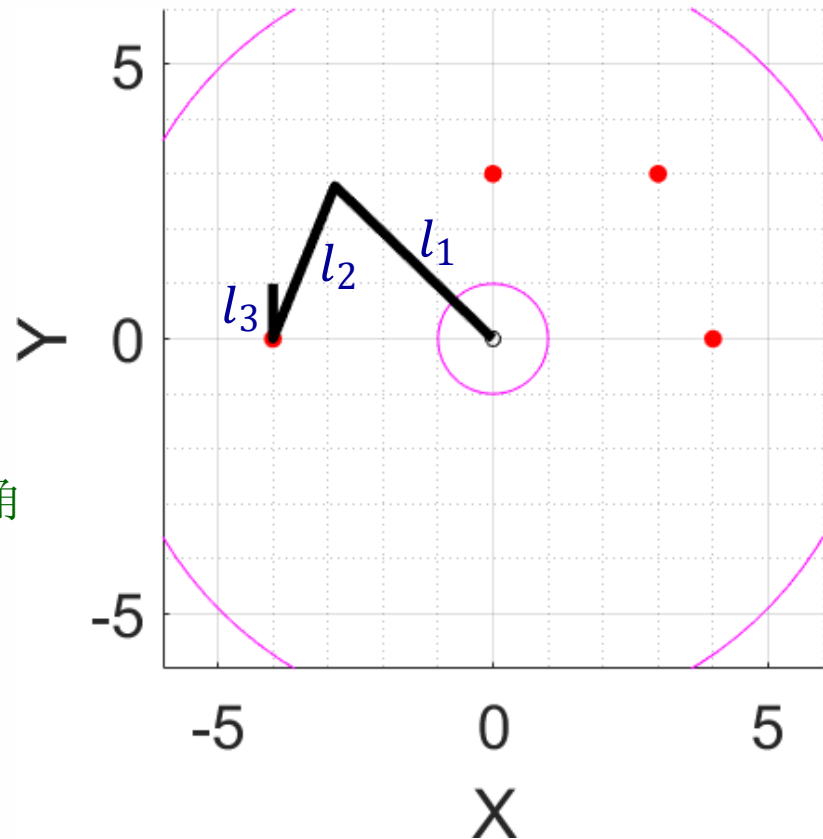
- 平面RRR手臂长度： $l_1 = 4$, $l_2 = 3$, and $l_3 = 1$

下表定义 initial, via, via, 和 final points的位置

t	x	y	theta
0	-4	0	90
2	0	3	45
4	3	3	30
7	4	0	0

(X,Y) 定义在第二杆件的末端

Theta为第三杆件对X坐标轴的夹角



示例：RRR机械臂-2

□ 方法一：以三次多项式在笛卡尔空间下规划轨迹

1. 求出3个DOF (X, Y, θ)各自三次多项式的系数

需通过4个点：每个DOF有3个三次多项式，共12个未知数

$$\Theta_{12 \times 1} = T_{12 \times 12} A_{12 \times 1}$$

为 $(\Delta t_1, \Delta t_2, \Delta t_3)$ 函数

X/Y/ θ

$$\begin{array}{l} \text{6 position} \\ \text{B.C.s} \\ \\ \text{2 init., final} \\ \text{vel.} \\ \\ \text{4 vel. \& acl.} \\ \text{continuity} \end{array} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_1 \\ \theta_2 \\ \theta_2 \\ \theta_f \\ \dot{\theta}_0 \\ \dot{\theta}_f \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & \Delta t_1 & \Delta t_1^2 & \Delta t_1^3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \Delta t_2 & \Delta t_2^2 & \Delta t_2^3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \Delta t_3 & \Delta t_3^2 & \Delta t_3^3 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2\Delta t_3 & 3\Delta t_3^2 \\ 0 & 1 & 2\Delta t_1 & 3\Delta t_1^2 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 6\Delta t_1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2\Delta t_2 & 3\Delta t_2^2 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 6\Delta t_2 & 0 & 0 & -2 & 0 \end{bmatrix} \begin{bmatrix} a_{10} \\ a_{11} \\ a_{12} \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ a_{31} \\ a_{32} \\ a_{33} \end{bmatrix}$$

示例：RRR机械臂-3

□ 方法一：以三次多项式在笛卡尔空间下规划轨迹

将三个DOF在各点的参数代入matlab后将会是：

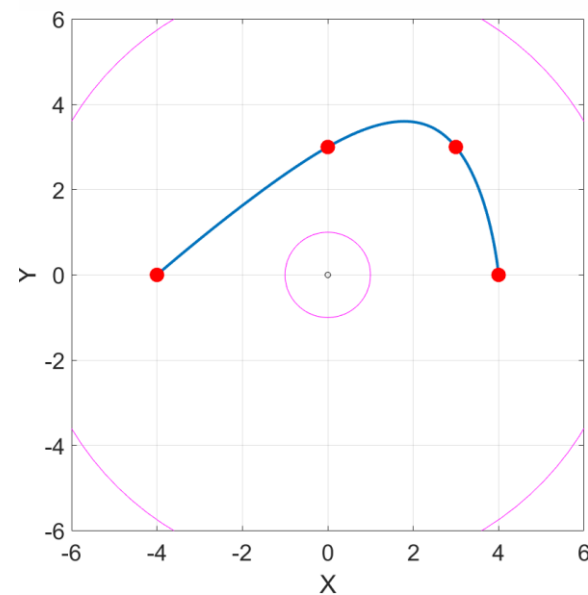
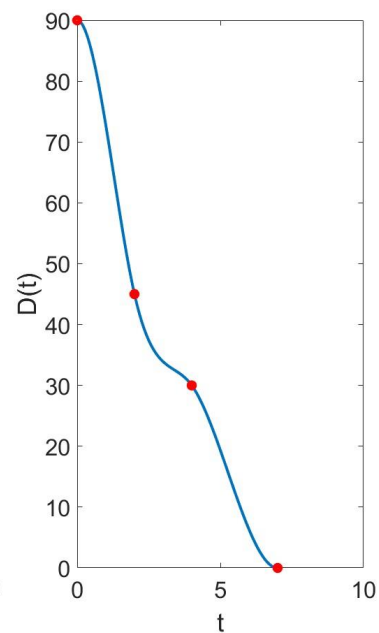
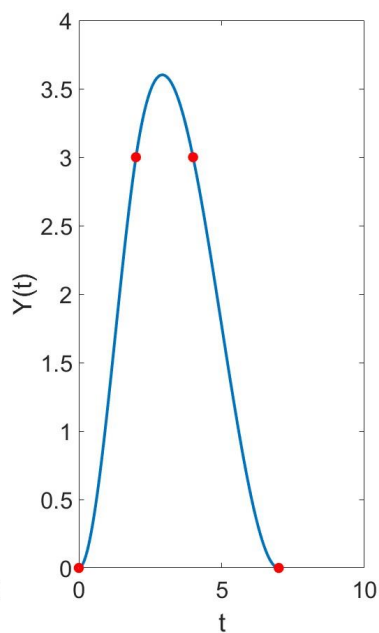
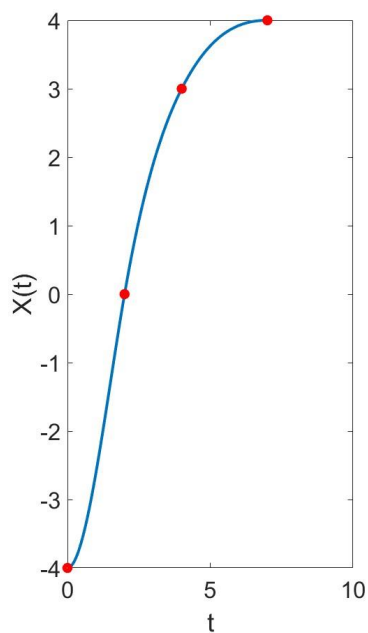
$$\theta_{12 \times 3} = T_{12 \times 12} A_{12 \times 3}$$

$$\begin{matrix} X & Y & \theta \end{matrix} \begin{bmatrix} -4 & 0 & 90 \\ 0 & 3 & 45 \\ 0 & 3 & 45 \\ 3 & 3 & 30 \\ 3 & 3 & 30 \\ 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 4 & 8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 3 & 9 & 27 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 6 & 27 & 0 \\ 0 & 1 & 4 & 12 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 12 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 4 & 12 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 12 & 0 & 0 & -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} A_{12 \times 3} \end{bmatrix}$$

示例：RRR机械臂-4

2. 对所有DOF规划平滑轨迹

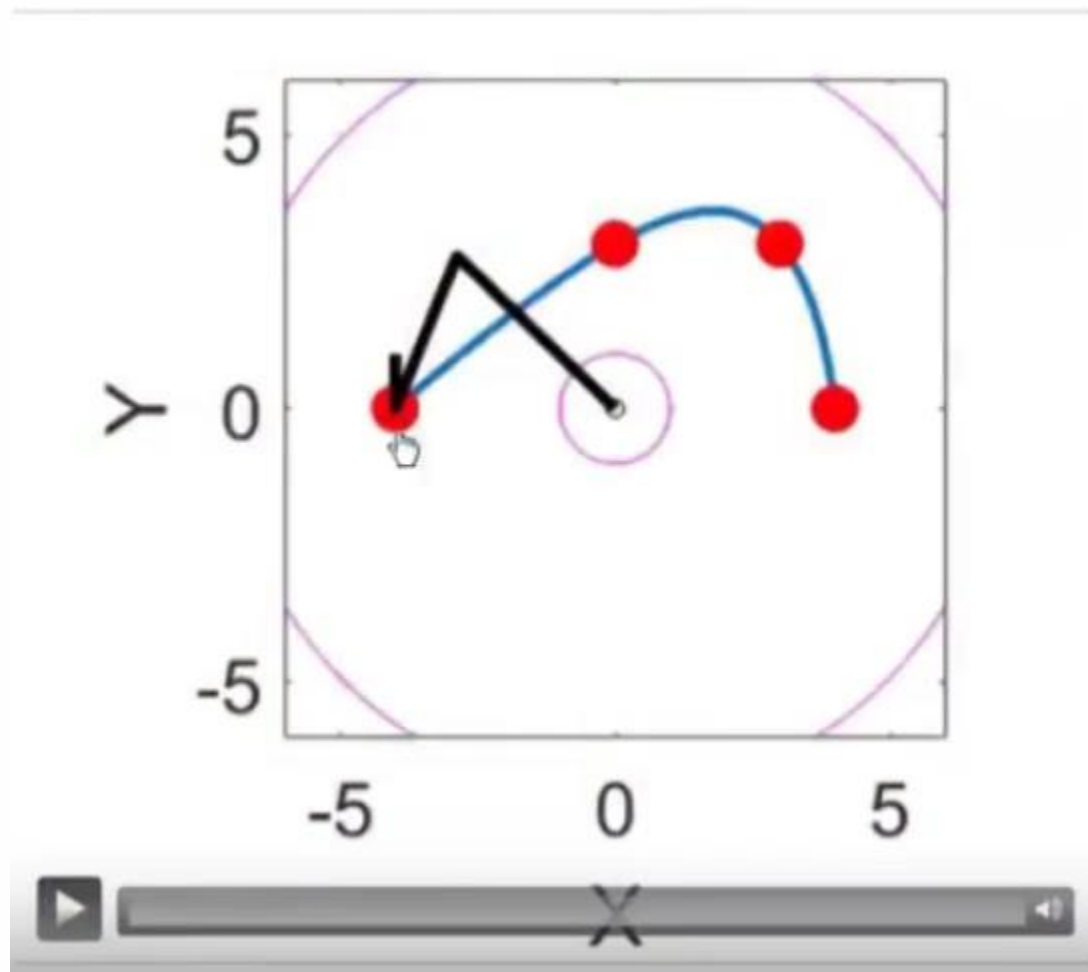
t	x	y	theta
0	-4	0	90
2	0	3	45
4	3	3	30
7	4	0	0



示例：RRR机械臂-5

3. I.K., 找出各关节的对应轨迹

4. 将关节带入手臂模拟动作，确认手臂末端点轨迹如规划运作



示例：RRR机械臂-6

□ 方法二：以三次多项式在关节空间下规划轨迹

1. I.K., 求出initial、via、final points的关节 angles ($\theta_1, \theta_2, \theta_3$)

2. 求出各($\theta_1, \theta_2, \theta_3$) 三次多项式的系数

需通过4个点：每个关节 angle有3个三次多项式，共12个未知数

$$\Theta_{12 \times 3} = T_{12 \times 12} A_{12 \times 3}$$

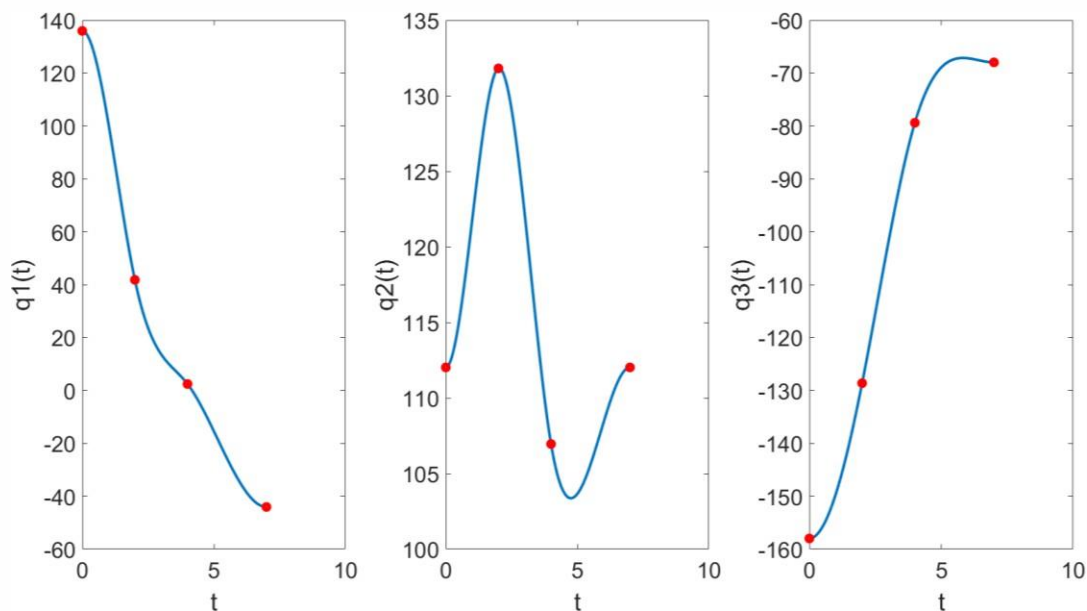
4 vel. & acl. continuity 6 position B.C.s

$$\begin{bmatrix} \theta_1 & \theta_2 & \theta_3 \end{bmatrix} \begin{bmatrix} 2.3728 & 1.9552 & -2.7572 \\ 0.7297 & 2.3005 & -2.2449 \\ 0.7297 & 2.3005 & -2.2449 \\ 0.0426 & 1.8668 & -1.3858 \\ 0.0426 & 1.8668 & -1.3858 \\ -0.7688 & 1.9552 & -1.1864 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 4 & 8 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 3 & 9 & 27 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 6 & 27 \\ 0 & 1 & 4 & 12 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 12 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 4 & 12 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 12 & 0 & 0 & -2 & 0 \end{bmatrix} \begin{bmatrix} A_{12 \times 3} \end{bmatrix}$$

示例：RRR机械臂-7

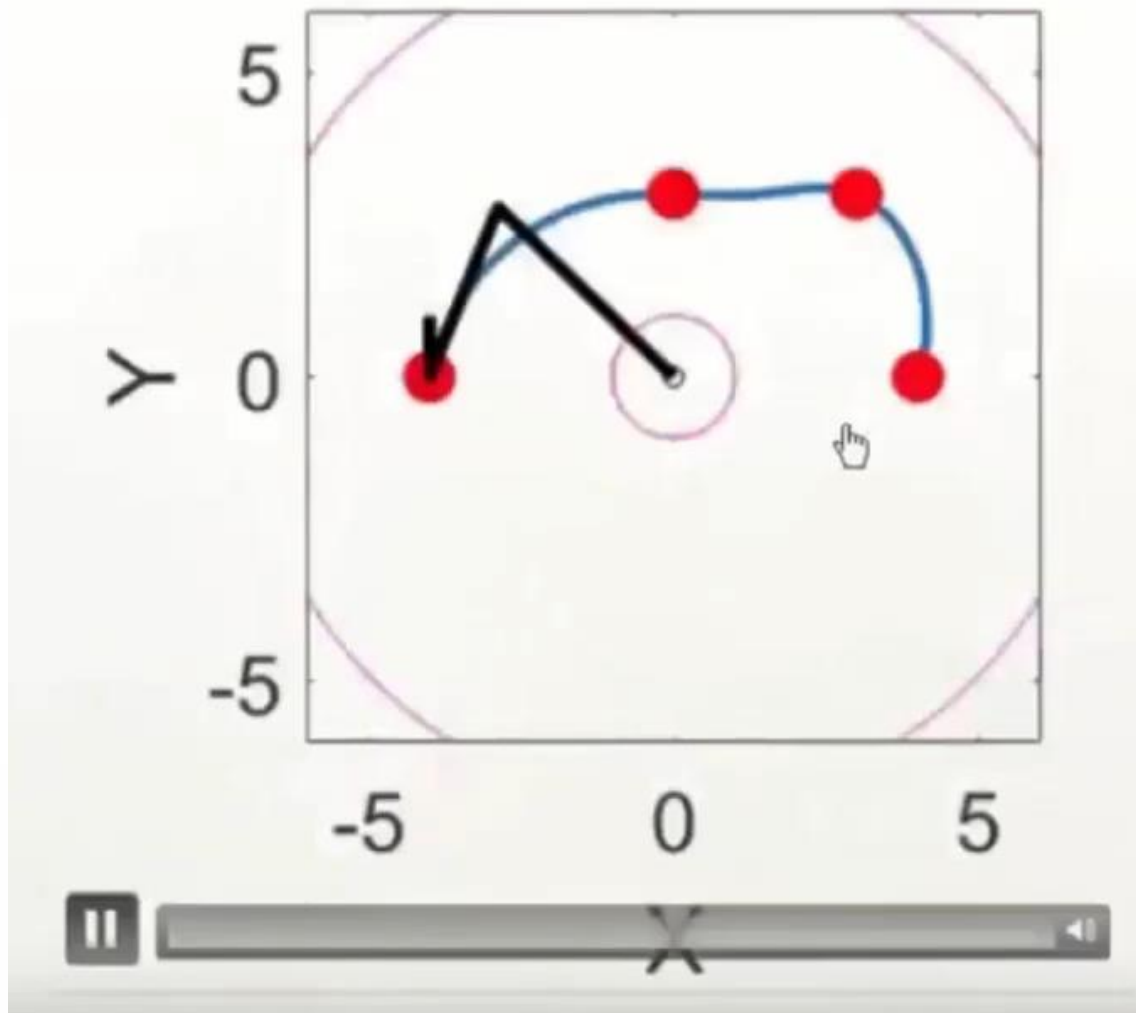
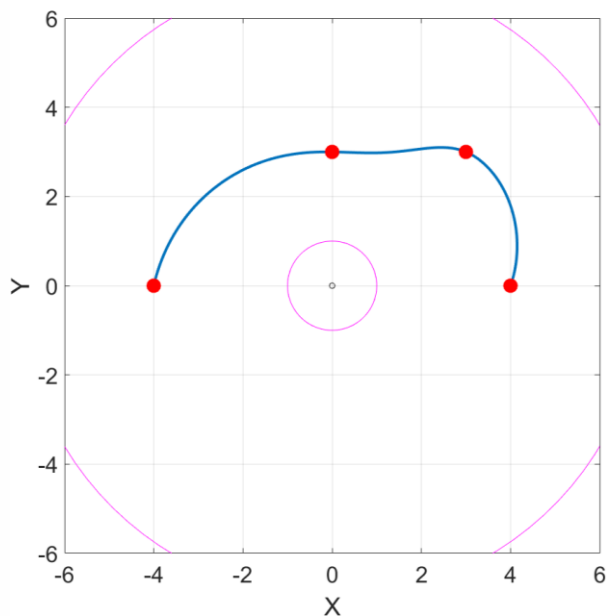
3. 对所有DOF规划平滑轨迹

t	x	y	theta
0	-4	0	90
2	0	3	45
4	3	3	30
7	4	0	0



示例：RRR机械臂-8

4. 将关节带入手臂模拟动作，确认手臂末端点轨迹如规划运作



High-order Polynomials

- 如果位置、速度、和加速度都必须规划

5

⇒ Quintic polynomial $\theta(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5 = \sum_{i=0}^5 a_i t^i$

有6个条件，quintic polynomial有6个未知数

$$\left\{ \begin{array}{l} \theta_0 = a_0 \\ \theta_f = a_0 + a_1\Delta t + a_2\Delta t^2 + a_3\Delta t^3 + a_4\Delta t^4 + a_5\Delta t^5 \\ \dot{\theta}_0 = a_1 \\ \dot{\theta}_f = a_1 + 2a_2\Delta t + 3a_3\Delta t^2 + 4a_4\Delta t^3 + 5a_5\Delta t^4 \\ \ddot{\theta}_0 = 2a_2 \\ \ddot{\theta}_f = 2a_2 + 6a_3\Delta t + 12a_4\Delta t^2 + 20a_5\Delta t^3 \end{array} \right.$$

$$a_0 = \theta_0 \quad a_3 = \frac{20(\theta_f - \theta_0) - (8\dot{\theta}_f + 12\dot{\theta}_0)\Delta t - (3\ddot{\theta}_0 - \ddot{\theta}_f)\Delta t^2}{2\Delta t^3}$$

$$a_1 = \dot{\theta}_0 \quad a_4 = \frac{30(\theta_0 - \theta_f) + (14\dot{\theta}_f + 16\dot{\theta}_0)\Delta t - (3\ddot{\theta}_0 - 2\ddot{\theta}_f)\Delta t^2}{2\Delta t^4}$$

$$a_2 = \frac{1}{2}\ddot{\theta}_0 \quad a_5 = \frac{12(\theta_f - \theta_0) - (6\dot{\theta}_f + 6\dot{\theta}_0)\Delta t - (\ddot{\theta}_0 - \ddot{\theta}_f)\Delta t^2}{2\Delta t^5}$$

1 linear

2 quadratic

3 cubic

4 quartic

5 quintic

6 hexic (sextic)

7 heptic (septic)

8 octic

9 nonic

10 decic